YINING ZHU

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PROFESSIONAL APPOINTMENTS

Duke University, Durham, NC *02/2026 – current*
- Postdoctoral Fellow with *Prof. John W. Hickey*

Johns Hopkins University, Baltimore, MD *10/2025 – 01/2026*
- Postdoctoral Associate with *Prof. Hai-Quan Mao*

EDUCATION

Johns Hopkins University, Baltimore, MD *07/2021 – 09/2025*
- Ph.D. in Biomedical Engineering

Duke University, Durham, NC *08/2024 – 09/2024*
- Visiting Research Scholar in Biomedical Engineering

Johns Hopkins University, Baltimore, MD *08/2019 – 05/2021*
- M.S.E. in Biomedical Engineering

Sichuan University, China *09/2015 – 06/2019*
- B.S. in Pharmacy

AWARDS & HONORS

Schmidt Science Fellows, *Schmidt Futures, US* *2026*
Forbes 30 Under 30 - Science, US *2026*
Siebel Scholar, *Siebel foundation, US* *2025*
Student Travel Achievement Recognition (STAR) award, *Society for Biomaterials, US* *2025*
Meritorious Abstract Travel Award, *American Society of Gene & Cell Therapy, US* *2024*
The Hans J. Prochaska Research Award, *Johns Hopkins University, US* *2024*
Student Travel Achievement Recognition (STAR) award, *Society for Biomaterials, US* *2023*
Outstanding Graduates Award in Sichuan Province, *Sichuan Province, China* *2019*
'Tang Lixin' Scholarship, *Sichuan University, China* *2018*
Top 100 Students Award, *Sichuan University, China* *2017*
Outstanding Chairman of the Student Union, *Sichuan University, China* *2017*
National Scholarship, *Ministry of Education of China* *2016; 2017; 2018*

PUBLICATIONS

*† Denotes equal contribution * Denotes corresponding author*
(34 peer-reviewed journal publications in total; Google Scholar)

First author publications:

14. **Y Zhu†,** C Wei†, *et al.*, HQ Mao*. Engineering age-adaptive mRNA lipid nanoparticle cancer vaccines via reprogramming systemic gene expression. ***(Nature Biomedical Engineering,*** *in revision*)

13. D Yu†, **Y Zhu†**, et al., HQ Mao*. Engineering lipid nanoparticles through integrated compositional and ligand targeting enhances β-cell–selective RNA delivery. (***Science Advances***, *submitted*)

12. X Liu†, **Y Zhu†**, WH Toh† et al., HQ Mao*. Engineering chemical topology in lipid nanoparticles for in vivo T-cell programming. (***Nature Nanotechnology***, *submitted*)

11. C Wei†, **Y Zhu†*,** *et al.*, SC Murphy*, HQ Mao*. Systemic trafficking of mRNA lipid nanoparticle vaccine following intramuscular injection generates potent tissue-specific T cell response. ***Nature Biomedical Engineering***, (2026). *(in press*)

10. X Liu†, **Y Zhu†,** *et al.*, HQ Mao*. Crosslinking of lipid nanoparticles enhances the delivery efficiency and efficacy of mRNA vaccines. ***Nature Chemical Engineering***, 3, 112–127, (2026).

9. J Lin†, **Y Zhu†,** *et al.*, HQ Mao*. Imidazolium lipid-based nanoparticles enable effective mRNA delivery and cellular immune response. ***Materials Today***, 95, 103265, (2026).

8. **Y Zhu†,** ZC Y†, S L†, *et al.*, HQ Mao*. An mRNA lipid nanoparticle-incorporated nanofiber-hydrogel composite for cancer immunotherapy. ***Nature Communications***, 16(1):5707, (2025).

7. X Lu†, **Y Zhu†,** *et al.*, HQ Mao*. A multi-step platform identifies spleen-tropic lipid nanoparticles for in vivo T cell-targeted delivery of gene editing proteins. ***Science Advances,*** 11(43):eady5579, (2025).
6. **Y Zhu**, *et al.*, SC Murphy*, HQ Mao*. Screening for lipid nanoparticles that modulate the immune activity of helper T cells towards enhanced antitumour activity. ***Nature Biomedical Engineering***, 8 (5), 544-560, (2024).
5. **Y Zhu†,** S Cai†, *et al.*, KW Leong*, HQ Mao*. Optimization of lipid nanoparticles for gene editing of the liver via intraduodenal delivery. ***Biomaterials***, 308:122559, (2024).
4. J Ma†, **Y Zhu†**, *et al.*, SX Sun*, HQ Mao*. Tuning extracellular fluid viscosity for enhanced transfection efficiency in genetic cell engineering. ***Nature Chemical Engineering***, 1(9): 576-587, (2024).
3. **Y Zhu**, *et al.*, SC Murphy*, HQ Mao*. Multi-step screening of DNA/lipid nanoparticles and co-delivery with siRNA to enhance and prolong gene expression. ***Nature Communications***, 13 (1), 4282, (2022).
2. Y Hu†, **Y Zhu†**, *et al.*, HQ Mao*. Size-controlled and shelf-stable DNA particles for production of lentiviral vectors. ***Nano Letters,*** 21 (13), 5697-5705, **(**2021).
1. **Y Zhu**, *et al.*, X Sun*. Albumin-biomineralized nanoparticles to synergize phototherapy and immunotherapy against melanoma. ***Journal of Controlled Release***, 322, 300-311, (2020).

Selective collaborative publications:

- B Torkzaban, **Y Zhu**, *et al.*, J Coller*. Use of poly adenosine tail mimetics to enhance mRNA expression from genes associated with haploinsufficiency disorders. ***Molecular Therapy Nucleic Acids***, 36 (1), 102453, (2025).
- L Cheng, **Y Zhu**, *et al.*, HQ Mao*. Machine learning elucidates design features of plasmid deoxyribonucleic acid lipid nanoparticles for cell type-preferential transfection. ***ACS nano***, 18 (42), 28735-28747, (2024).
- W Chen, **Y Zhu**, *et al.*, J He*. Potentiating the systemic immunity by bacteria-delivered STING activation in a tumor microenvironment. ***Advanced Functional Materials***, 33 (52), 2307001, (2023).
- Z Guo, **Y Zhu**, *et al.*, X Sun*. Rapid development of a subunit nano-vaccine against drug-resistant Pseudomonas aeruginosa with effective cross-protection. ***Nano Today***, 43, 101398, (2022).
- J Xue, **Y Zhu**, *et al.*, X Sun*. Nanoparticles with rough surface improve the therapeutic effect of photothermal immunotherapy against melanoma. ***Acta Pharmaceutica Sinica B***, 12 (6), 2934-2949, (2022).
- ZC Yao, YH Yang, J Kong, **Y Zhu**, *et al.*, HQ Mao*. Biostimulatory micro-fragmented nanofiber-hydrogel composite improves mesenchymal stem cell delivery and soft tissue remodeling. ***Small***, 18 (36), 2202309, (2022).

Reviews:

- W Chen, **Y Zhu**, *et al.*, X Sun*. Path towards mRNA delivery for cancer immunotherapy from bench to bedside. ***Theranostics***, 14 (1), 96, (2024).
- W Chen, **Y Zhu**, *et al.*, X Sun*. Advances in Salmonella Typhimurium-based drug delivery system for cancer therapy. ***Advanced Drug Delivery Reviews***, 185, 114295, (2022).
- S Huang, **Y Zhu**, *et al.*, Z Zhang*. Recent advances in delivery systems for genetic and other novel vaccines. ***Advanced Materials***, 34 (46), 2107946, (2022).

PATENTS

1. **Y Zhu**, X Liu, H Mao. Compositions and methods of preparing RNA lipid nanoparticles with enhanced stability and transfection efficiency. US Provisional Patent Application; Filed 7/19/2025.
2. **Y Zhu**, C Wei, H Mao. Lipid nanoparticles with integrated glycolipid adjuvant to promote tissue-specific cellular immunity. US Provisional Patent Application; Filed 2/18/2025.
3. **Y Zhu**, C Wei, D Yu, H Mao. Lipid nanoparticle formulations capable of migrating to systemic organs following intramuscular administration. US Provisional Patent Application; Filed 1/28/2025.
4. **Y Zhu**, X Lu, H Mao. Composition screening of lipid nanoparticle for intracellular delivery of gene-editing proteins. PCT/US2025/023531; Filed 4/7/2025.
5. **Y Zhu**, C Wei, J Ma, H Mao, et al. A mRNA lipid nanoparticle incorporated nanofiber-hydrogel composite to generate a local immunostimulatory niche for immunotherapy. PCT/US2025/023530; Filed 4/7/2025.
6. **Y Zhu**, J Ma, H Mao, et al. Composition of media with defined fluid viscosity for enhancing intracellular delivery of nanoparticles and viral vectors, and methods of use. PCT/US2024/039036; Filed 7/22/2024.
7. **Y Zhu**, H Mao, *et al.* Compositions of Lipid Nanoparticles for Plasmid DNA Delivery to the Liver and Methods for Preparing the Same. PCT/US2023/016938; Filed 3/30/2023.
8. **Y Zhu**, Y Hu, H Mao. Methods for preparation of plasmid DNA/lipid particles with defined size for in vitro and in vivo transfection. PCT/US2023, 18/546,221; Filed 8/11/2023.

9. **Y Zhu**, Y Hu, H Mao. Composition of shelf-stable plasmid DNA/PEI particles with defined sizes for virus production and method for preparation of the same. PCT/US2023, 18/546,222, Filed 8/11/2023.
10. **Y Zhu**, Y Hu, H Mao. Methods for preparation of shelf-stable plasmid DNA/polycation particles with defined sizes for cell transfection. PCT/US2022, 18/261,944; Filed 7/18/2023.

CONFERENCE PRESENTATIONS

1. **Zhu Y**, Ma J, *et al.*, Mao HQ. Enhancing Cell Transfection Efficiency via Modulation of Extracellular Fluid Viscosity. *Society of Biomaterials Annual Meeting and Exposition*. 2025. **Oral Presentation.**
2. **Zhu Y**, Yao Z-C, Li S, *et al.*, Mao HQ. mRNA lipid nanoparticle-incorporated nanofiber-hydrogel composite generates a local immunostimulatory niche for cancer immunotherapy. *Society of Biomaterials Annual Meeting and Exposition*. 2025. **Oral Presentation & Student Travel Achievement Recognition (STAR) award.**
3. **Zhu Y**, Yao Z-C, Li S, *et al.*, Mao HQ. Engineering A Biomaterials-based Lymphoid Niche for mRNA Lipid Nanoparticle Cancer Vaccines. *Biomedical Engineering Society Annual Meeting*. 2024. **Oral Presentation.**
4. **Zhu Y**, Yao Z-C, Li S, *et al.*, Mao HQ. A mRNA lipid nanoparticle incorporated nanofiber-hydrogel composite generates a local immunostimulatory niche for cancer immunotherapy. *American Society of Gene & Cell Therapy Annual Meeting*. 2024. **Oral Presentation & Meritorious Abstract Travel Award.**
5. **Zhu Y**, Ma J, Shen R, Vuong I, Mao HQ. Lipid Nanoparticle Composition Shapes Immune Response to mRNA Vaccine and Potency of Anticancer Immunity. *Society of Biomaterials Annual Meeting and Exposition*. 2023. **Oral Presentation & Student Travel Achievement Recognition (STAR) award.**
6. **Zhu Y**, Ma J, Shen R, Vuong I, Mao HQ. Compositional Optimization of mRNA Lipid Nanoparticles to Modulate Th1/Th2 Immune Activation Profile and Potentiate Anticancer Immunity. *American Society of Gene & Cell Therapy Annual Meeting*. 2023. **Poster Presentation.**
7. **Zhu Y**, Shen R, Vuong I, Hu Y, Mao HQ. Multi-step Screening and Composition Optimization of Lipid Nanoparticles for Liver-targeted Plasmid DNA Delivery. *Society of Biomaterials Annual Meeting and Exposition*. 2022. **Oral Presentation**.

PROFESSIONAL MEMBERSHIPS

- **Society for Biomaterials** 2021 – Present
- **American Society of Gene & Cell Therapy** 2021 – Present
- **Biomedical Engineering Society** 2024 – Present

TEACHING EXPERIENCE

- **EN.580.109.12 | Introduction to Nanomedicine** 01/2023, 01/2024

  JHU *Intersession Course Instructor*

  Introduced and offered a comprehensive view of nanomedicine, including the physical and chemical basis of biomaterials in the nano-size range, bio-interactions governing efficacy and side effects, conventional and advanced design strategies to overcome biological barriers, and examples in diverse applications.
- **EN.580.453 | Immunoengineering: Principles and Applications** 09/2024 – 12/2024

  JHU *Teaching Assistant*

  Offered insightful explanations and practical demonstrations in class, fostering a collaborative learning environment that supported students in mastering the fundamental principles of immunoengineering.
- **EN.580.642 | Tissue Engineering** 09/2022 – 12/2022

  JHU *Teaching Assistant*

  Provided insightful explanations and practical demonstrations in classes. Helped create a collaborative learning environment to assist students in learning the fundamental principles in tissue engineering.

LEADERSHIP & PROFESSIONAL SERVICES

- **Lab Manager**, Mao Laboratory, Johns Hopkins University 2021/09 – 2025/05
- **Peer Health Navigators**, Johns Hopkins University 2023/09 – 2024/09
- **Intern Pharmacist**, West China Hospital, Sichuan University 2018/07 – 2018/08
- **President, Student Union of West China School of Pharmacy**, Sichuan University 2015/05 – 2018/05
- **Journal Reviewer**
  - Serve as reviewers for *Biomaterials, Journal of Controlled Release, Cancer Nanotechnology, iScience, Pharmaceutical Research, Bioengineering & Translational Medicine, Journal of Drug Delivery Science and Technology, Journal of Liposome Research, Scientific Reports, Discover Chemistry, Journal of Nanobiotechnology*.